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Clinical Research Article

Machine Learning–Based Biomarker Identification for Early Diagnosis of Metabolic Dysfunction–Associated Steatotic Liver Disease

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Abstract

Context: Metabolic dysfunction–associated steatotic liver disease (MASLD) is an umbrella term for simple hepatic steatosis and the more severe metabolic dysfunction–associated steatohepatitis. The current reliance on liver biopsy for diagnosis and a lack of validated biomarkers are major factors contributing to the overall burden of MASLD.

Objective: This study investigates the association between biomarkers and hepatic steatosis and stiffness measurements, measured by FibroScan®.

Methods: Data from the National Health and Nutritional Examination Survey (2017-2020) were collected for 15 560 patients. Propensity score matching balanced the data with a 1:1 case to control for age and sex allowing for preliminary trend assessment. Random Forest machine learning determined variable importance for the incorporation of key biomarkers (age, sex, race, BMI, HbA1c, plasma fasting glucose, insulin, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, ALT, AST, ALP, albumin, GGT, LDH, iron, total bilirubin, total protein, uric acid, BUN, and hs-CRP) into logistic regression models predicting steatosis (MASLD indicated by a controlled attenuation parameter score of ≥238 dB/m) and stiffness (hepatic fibrosis indicated by a median liver stiffness ≥7 kPa). Sensitivity analysis using XGBoost and Recursive Feature Elimination was performed.

Results: The Random Forest models (the most accurate) predicted MASLD with 79.59% accuracy ($P < .001$) and specificity of 84.65% and predicted hepatic fibrosis with 86.07% accuracy ($P < .001$) and sensitivity of 98.01%. Both the steatosis and stiffness models identified statistically significant biomarkers, with age, BMI, and insulin appearing significant to both.

Conclusion: These findings indicate that assessing a variety of biomarkers, across demographic, metabolic, lipid, and standard biochemistry categories, may provide valuable initial insights for diagnosing patients for MASLD and hepatic fibrosis.

Key Words: MASLD, hepatic fibrosis, biomarker

Abbreviations: ALF, American Liver Foundation; ALP, alkaline phosphatase; ACTH, adrenocorticotropic hormone; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; BUN, blood urea nitrogen; CAP, controlled attenuation parameter; CVD, cardiovascular disease; DSI, Dallas Steatosis Index; GGT, gamma-glutamyl transferase; HbA1c, glycated hemoglobin; HDL, high-density lipoprotein; hs-CRP, high-sensitivity C-reactive protein; %IncMSE, percent increase in mean-squared error; LDL, low-density lipoprotein; MASH, metabolic dysfunction–associated steatohepatitis; MASLD, metabolic dysfunction–associated steatotic liver disease; MetS, metabolic syndrome; NHANES, National Health and Nutritional Examination Survey; PFG, plasma fasting glucose; UA, uric acid; VIF, variance inflation factor.

Metabolic dysfunction–associated steatotic liver disease (MASLD) is the most common chronic liver disorder worldwide, presenting as spectrum of liver disease beginning with steatosis and progressing to steatohepatitis, cirrhosis, and sometimes hepatocellular carcinoma (1). Although MASLD is the most rapidly increasing cause of mortality associated with the liver, it has led to significant disease burden due to its growing prevalence and close association with metabolic syndrome (MetS: a collection of metabolic conditions, including obesity, type 2 diabetes mellitus, insulin resistance, dyslipidemia, and hypertension) and cardiovascular disease (CVD) (1-4).

MASLD has been regarded as the hepatic manifestation of MetS, suggesting similar pathophysiological processes. The prevalence of MetS risk factors, especially obesity and type

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2 diabetes, continues to rise concomitantly with MASLD (2, 3). The American Liver Foundation (ALF) estimates MASLD prevalence at approximately 25%, although that might be an underestimation, with one study estimating a prevalence of around 31.9% (5, 6). A current ALF public health initiative, Think Liver Think Life®, has produced preliminary data indicating MASLD in about 64% of screened patients, with an upcoming study by the federal government aiming to reveal the true prevalence of MASLD nationwide (7). Liver biopsy has traditionally been the gold standard for the diagnosis of MASLD and its disease states. However, it carries increased risks due to its invasive nature and inapplicability to a large scale (8). The last decade has seen an advancement in promising alternatives to liver biopsy, including noninvasive imaging modalities, such as transient elastography (FibroScan®). With these imaging tools, repeat testing is safer and easier to conduct, allowing for earlier disease detection and more frequent monitoring (9, 10).

Although noninvasive imaging modalities have improved detection of MASLD and metabolic dysfunction–associated steatohepatitis (MASH), they have limitations such as operator dependence, cost of additional tests, and the potential to produce altered results when encountering certain patient characteristics. Hence, the identification of accurate and reliable biomarkers remains a critical research goal (8, 9, 11). Our study investigates the association between various biomarkers (related to demographics, metabolic risk, lipid components, and standard biochemical indices) and hepatic steatosis and stiffness (measured through FibroScan) in a large population of patients surveyed by the National Health and Nutritional Examination Survey (NHANES). Utilizing machine learning and logistic regression models, we aim to identify biomarker profiles predictive of MASLD and hepatic fibrosis to aid early diagnosis and risk assessment in clinical practice.

Methods

Designed by the National Center for Health Statistics (NCHS), the NHANES gathers health and nutritional information from US adults and children through medical interviews and physical examinations. Survey data is publicly available to assess disease states and risk factors nationally and, in turn, initiate improvements to population health. The NHANES 2017-March 2020 Data was the dataset used in this study, including demographic, socioeconomic, health interview, and physical examination variables (12, 13). The programming language R was used for data analysis of an initial sample of 15 560 participants (14, 15). Participants with hepatitis B, hepatitis C, and daily or nearly daily alcohol drinkers were identified and excluded from the study. Participants with hepatitis B and hepatitis C were identified using questionnaire data assessing hepatitis status (16). Those who drank alcohol every day or nearly every day were determined by using an alcohol use questionnaire (17). All the other participants were included in the study, resulting in a final sample size of 12 471 participants.

Based on literature review, pertinent variables from these data subsets were selected and merged using R (18-25). A comprehensive set of variables was chosen based on clinical relevance to MASLD and its risk factors or associations in previous research (20, 22, 26-28). Hepatic steatosis and hepatic stiffness were measured using liver ultrasound transient elastography (FibroScan). This technology is based on the principle that normal liver parenchyma is soft and flexible, whereas fibrotic liver is stiff. Hepatic steatosis causes ultrasound attenuation, which is measured by controlled attenuation parameter (CAP™) as a marker of fatness of the liver (29) In FibroScan, a vibration of mild amplitude and low frequency (50 Hz) is transmitted through the skin by a vibrating probe tip placed in the intercostal space. A shear wave is produced and travels through the liver (29). The velocity of the shear wave is directly proportional to liver stiffness (fibrosis), which is measured in kilopascals (kPa) (29). Hepatic steatosis, evaluated using CAP, is measured simultaneously with stiffness measurements. Quality control and assurance measures were also adopted, including regular training of NHANES health technicians by a FibroScan technician along with regular data reviews of readings and site visits by a senior medical officer (29). Each health technician recorded 10 raw data measurements from which interquartile range (IQR) and median stiffness (M) were calculated (29). An IQR/M ratio for the 10 measurements was then calculated. Measurements were deemed satisfactory when the ratio was <30% to reduce variability and outliers (29).

Participants with hepatic steatosis (CAP score of ≥238 dB/m) were identified as the MASLD group due to the disease's main characterization of ectopic fat accumulation. Participants with hepatic stiffness (median liver stiffness ≥7 kPa) were defined as having hepatic fibrosis, as this state is characterized by scarring of liver tissue (30).

To balance the data, propensity score matching was performed for the outcomes: steatosis and stiffness. Due to the requirement of complete data for propensity score matching, the sample sizes decreased significantly for the steatosis (n = 2714) and stiffness (n = 988) groups. Age and sex were adjusted for, and case and control were matched 1:1 for the selected demographic, metabolic, lipid, and standard biochemistry profile variables to determine their relationship with steatosis and stiffness. To determine metabolic effects on MASLD and hepatic fibrosis, body mass index (BMI), percent glycohemoglobin (HbA1c), insulin, and plasma fasting glucose (PFG) were analyzed. To assess the effect of various lipid profile components, triglyceride levels, total cholesterol, low-density lipoprotein (LDL) cholesterol, and high-density lipoprotein (HDL) cholesterol data were compared. To investigate the relationship of standard biochemistry profile values with MASLD/hepatic fibrosis, liver enzymes (alanine aminotransferase [ALT]/ aspartate aminotransferase [AST]/ alkaline phosphatase [ALP]/ gamma-glutamyl transferase [GGT]), albumin, blood urea nitrogen (BUN), uric acid (UA), total bilirubin, total protein, and iron were studied. To identify socioeconomic risk factors, income to poverty ratio was studied. The mean value of each potential biomarker was compared in participants with MASLD (≥238 dB/m) vs without hepatic steatosis, and in participants with hepatic fibrosis (≥7 kPa) vs without hepatic stiffness in a binary comparison.

Based on findings from our propensity score matching analysis, a thorough literature review, and clinical accessibility, potential biomarker variables (Table 1) and the 2 outcome variables (binary steatosis and stiffness outcomes) were selected for logistic regression models. For maximal sample size, the data were unmatched. Any entries with missing data were omitted at this step, resulting in data from 3857 patients being included in each model. Random Forest machine learning algorithm (31) determined variable importance by percent increase in mean-squared error (%IncMSE). The potential

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Table 1. List of variables selected for possible inclusion in hepatic steatosis and stiffness logistic regression models

- Age (years)[a]
- Alanine aminotransferase (ALT) (U/L)
- Albumin, refrigerated serum (g/dL)
- Alkaline phosphatase (ALP) (IU/L)
- Aspartate aminotransferase (AST) (U/L)
- Blood urea nitrogen (mg/dL)
- BMI (kg/m^2)
- Direct HDL-cholesterol (mmol/L)
- Gamma-glutamyl transferase (GGT) (IU/L)
- Glycohemoglobin (A1c) (%)
- High-sensitivity C-reactive protein (hs-CRP) (mg/L)
- Insulin (μU/mL)
- Iron, refrigerated serum (μg/dL)
- Lactate dehydrogenase (LDH) (IU/L)
- LDL-cholesterol, Friedewald (mg/dL)
- Plasma fasting glucose (PFG) (mg/dL)
- Race/ethnicity
- Sex
- Total bilirubin (mg/dL)
- Total cholesterol (mg/dL)
- Total protein (g/dL)
- Triglycerides (mg/dL)
- Uric acid (mg/dL)

[a]Age is topcoded at 80 years of age by NHANES.

biomarker variables were integrated into each logistic regression model. The data were split for each model with 80% of data used for model building and 20% used for testing the model's prediction accuracy, and the testing data included 771 patients for both steatosis and stiffness models. The reference for sex was female and for race was Mexican American, which were chosen automatically by the model. Both steatosis and stiffness logistic regression models were then converted into confusion matrices curves to be further analyzed.

As a sensitivity analysis, feature selection was also performed using XGBoost (32) and Recursive Feature Elimination (33) followed by logistic regression for both steatosis and stiffness using the most important variables. To address multicollinearity in our multivariate logistic regression models, the variance inflation factor (VIF) was determined post hoc for each variable in both stiffness and steatosis models. A flowchart describing data collection method, feature selection, and statistical analyses performed for the study is available in Fig. 1.

Results

Patient Demographics

Our initial patient population consisted of 12 471 patients included in the 2017-2020 Pre-Pandemic NHANES dataset. Of these patients, 5266 were included in the MASLD group with hepatic steatosis on transient elastography exam while 1343 were members of the hepatic fibrosis group with hepatic stiffness on FibroScan. The average age of our total patient population ($n = 12\,471$) was 38.34 years. However, patients in the MASLD ($n = 5266$) and hepatic fibrosis ($n = 1343$) groups were older, with an average age of 48.16 and 50.36 years, respectively (Table 2). The sex distribution of the total patient population and the MASLD group was almost even among male and female individuals (48.3% vs 51.7% and 50.9% vs 49.1%, respectively) while the hepatic fibrosis patient group exhibited a greater proportion of male individuals (56.4%).

Non-Hispanic White patients displayed the greatest proportion of individuals surveyed, at 33.0% in the total population. This trend continued in MASLD and hepatic fibrosis groups, where the Non-Hispanic White patients comprised 33.6% and 34.9% of the groups, respectively. Non-Hispanic Black patients made up 26.4% of the total survey population, yet this proportion decreased for the MASLD group (23.3%) and increased in the hepatic fibrosis group (30.0%). The Non-Hispanic Asian group made up 11.6% of the total population, similar to that in the MASLD group (11.7%) and decreased in the hepatic fibrosis group (7.1%). Although only 13.0% of the total population identified as Mexican American, a greater proportion was displayed in both MASLD (15.1%) and hepatic fibrosis (13.6%) patient groups. Although only a small proportion of the total population sampled identified as a part of the Other Hispanic (10.0%) and Other Race/Multi-Racial (6.1%) groups, there was a slight increase for the proportion of Other Hispanic patients in the MASLD group (11.0%) and a marginal decrease in the hepatic fibrosis group (9.5%). Similarly, the proportion of Other Race/Multi-Racial patients changed minimally among MASLD (5.3%) and hepatic fibrosis (4.8%) patient groups (Table 2). Among the total, MASLD, and hepatic fibrosis groups, the ratios of income to poverty with the largest proportion were "0-1," "1.01-2" and "4.01-5," indicating a concentration of participants at the poorest and wealthiest ratios (Table 2). Thus, patients with MASLD and hepatic fibrosis were older, and more likely of male sex, of Non-Hispanic White origin, and belonging to the extremes of income spectrum.

Propensity Score Matching

In the steatosis groups, all potential biomarkers produced a statistically significant P value ($P < .05$) upon a t test between the means for the participants with steatosis (MASLD) and without steatosis (normal) except for iron ($P = .570$) and bilirubin ($P = .417$). Additionally, the mean age varied widely among groups, with the average age of the MASLD group being much higher at 58.78 years and the normal group at 36.08 years ($P < .001$). Elevated mean values were found in the MASLD group for all 4 metabolic profile variables (HbA1c, PFG, insulin and BMI) with strong statistical significance (HbA1c, $P < .001$; PFG, $P < .001$; Insulin, $P < .001$; and BMI: $P < .001$). Of the variables associated with the lipid profile, triglycerides produced the most statistically significant P value ($P < .001$) and had a mean of 123.92 mg/dL in the MASLD group vs 74.37 mg/dL in the normal group. Lastly, of the standard biochemistry profile variables producing a statistically significant P value between the 2 groups, uric acid indicated the strongest statistical significance with a P value of <.001. The mean was 5.70 mg/dL in the MASLD group and 4.98 in the normal group (Table 3).

In the stiffness groups, only BUN, total bilirubin, and total protein were not statistically significant between participants

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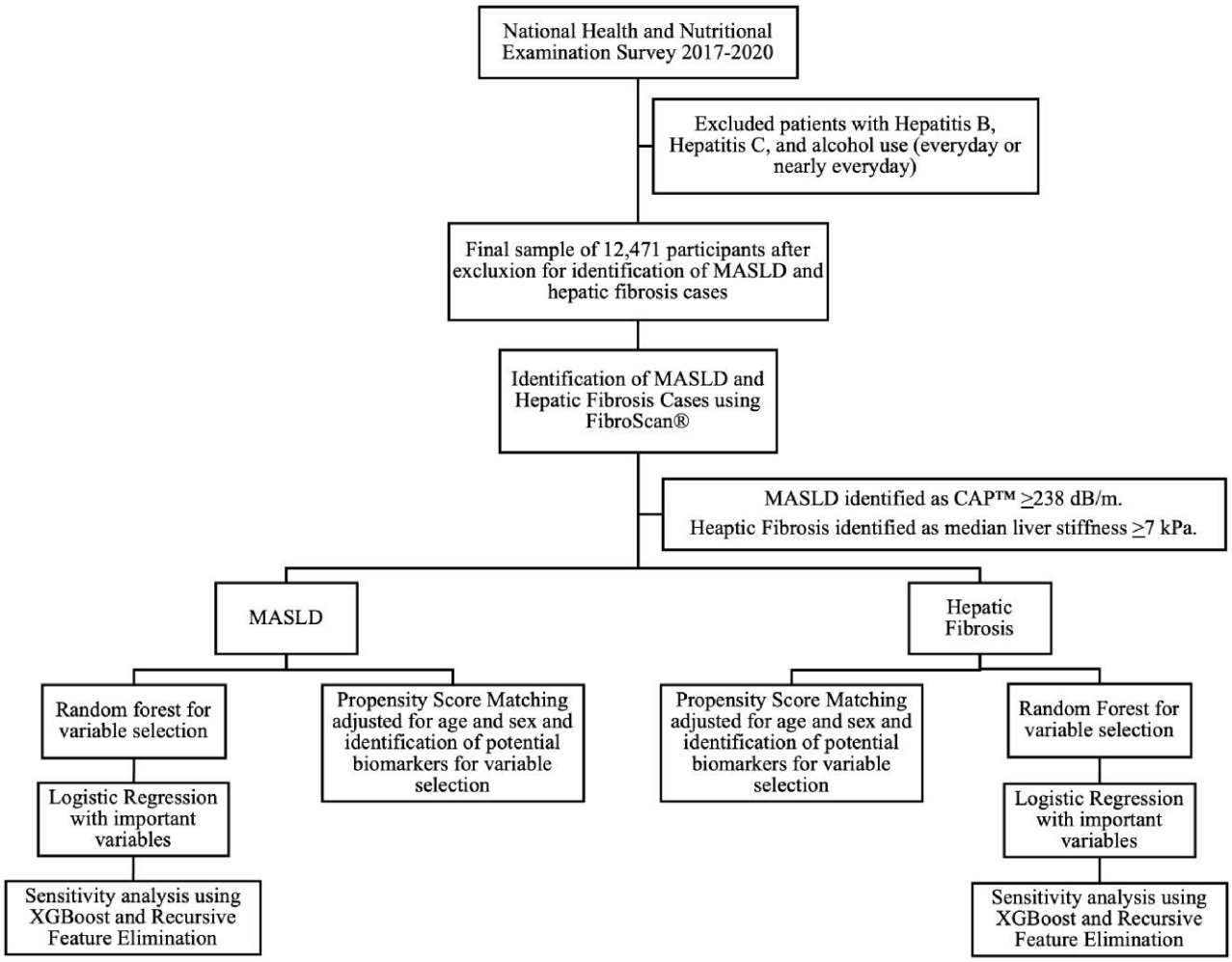


Figure 1. Flowchart explaining data collection, feature selection, and statistical analyses.

with and without hepatic fibrosis ($P = .051$, .940, .411, respectively). However, the matched data indicated the same age and sex distribution for the stiffness groups, unlike the steatosis comparison, and the sample size was reduced ($n = 988$). Although the steatosis group displayed even smaller P values for the metabolic profile variables, HbA1c, PFG, insulin and BMI again showed strong statistical significance between people with and without hepatic fibrosis (A1c: $P < .001$, PFG: $P < .001$, insulin: $P < .001$, and BMI: $P < .001$). Similarly, the P-values were decreased in the group with hepatic fibrosis compared to normal for the lipid profile with, HDL-cholesterol showing the greatest statistically significant difference between means with an average of 48.30 mmol/L and 54.18 mmol/L, respectively ($P < .001$). Total cholesterol and LDL-cholesterol showed a decrease in the hepatic fibrosis group compared to participants without the disease ($P < .001$) which contrasts with that seen in MASLD group where it increased compared to normal ($P < .001$). Of the standard biochemistry profile variables, the smallest P value ($P < .001$) was associated with ALT, where the average for participants with hepatic fibrosis was 27.11 U/L, and the average for people without disease was 20.48 U/L. The results are available in Table 3.

Overall, there was statistically significant variability among participants with disease (MASLD; hepatic fibrosis) and without disease for most potential biomarkers. Differences in the sample size among steatosis and stiffness groups as well as the matching process are present.

Logistic Regression Models

Using logistic regression models and Random Forest machine learning (34), variable importance analysis for the stiffness outcome resulted in the exclusion of variables having the lowest %IncMSE values. The variables included in the stiffness model were age, sex, BMI, insulin, HbA1c, PFG, LDL, total cholesterol, ALT, AST, BUN, GGT, and high-sensitivity C-reactive protein (hs-CRP). The variables included in the steatosis model were age, race, sex, BMI, HbA1c, insulin, PFG, LDL, HDL, total cholesterol, triglycerides, ALT, AST, ALP, albumin, BUN, GGT, iron, lactate dehydrogenase (LDH), total bilirubin, UA, and hs-CRP.

People with hepatic stiffness were more likely (OR = 1.008, $P = .023$) to be older. Female patients were 34.9% less likely ($P = .001$) than male patients to have hepatic stiffness. Metabolic profile variables were significantly elevated for hepatic stiffness except PFG and HbA1c, with BMI showing the strongest significance (OR = 1.106, $P < .001$). Lipid profile variables were not significant for hepatic stiffness. Among the standard biochemistry profile variables in hepatic stiffness, AST (1.7%, $P = .007$), BUN (2.4%, $P = .017$), and GGT (0.6%, $P < .001$) were more likely to be elevated with the

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Table 2. Patient demographics

	Total	Steatosis			Stiffness		
		With hepatic steatosis (MASLD)	Without hepatic steatosis (normal)	*P* value	With hepatic stiffness (hepatic fibrosis)	Without hepatic stiffness (normal)	*P* value
Age (years)							
N	12 471	5266[c]	3738[c]	<.001[a]	1343[c]	7663[c]	<.001[a]
Mean	38.34	48.16	36.62		50.36	42.14	
Median	36	50	31		54	41	
Range	6.00, 80.00[a]	12.00, 80.00[a]	12.00, 80.00[a]		12.00, 80.00[a]	12.00, 80.00[a]	
Sex, n (%)							
Male	6019 (48.3%)	2682 (50.9%)	1701 (45.5%)	<.001[b]	758 (56.4%)	3626 (47.3%)	<.001[b]
Female	6452 (51.7%)	2584 (49.1%)	2037 (54.5%)		585 (43.6%)	4037 (52.7%)	
Race, n (%)							
Mexican American	1621 (13.0%)	794 (15.1%)	367 (9.8%)	<.001[b]	183 (13.6%)	978 (12.8%)	<.001[b]
Other Hispanic	1243 (10.0%)	578 (11.0%)	356 (9.5%)		128 (9.5%)	806 (10.5%)	
Non-Hispanic White	4112 (33.0%)	1770 (33.6%)	1207 (32.3%)		469 (34.9%)	2508 (32.7%)	
Non-Hispanic Black	3288 (26.4%)	1226 (23.3%)	1120 (30.0%)		403 (30.0%)	1945 (25.4%)	
Non-Hispanic Asian	1441 (11.6%)	618 (11.7%)	465 (12.4%)		96 (7.1%)	987 (12.9%)	
Other Race/Multi-Racial	766 (6.1%)	280 (5.3%)	223 (6.0%)		64 (4.8%)	439 (5.7%)	

Income to Poverty Ratio	Total (N = 10 651)	With hepatic steatosis (MASLD) (N = 7792)	Without hepatic steatosis (Normal)	*P* value	With hepatic stiffness (hepatic fibrosis) (N = 7794)	Without hepatic stiffness (normal)	*P* value
0-1	2407 (22.6%)	880 (19.4%)	753 (23.1%)	.001[b]	233 (19.9%)	1400 (21.1%)	.013[b]
1.01-2	2868 (26.9%)	1258 (27.7%)	839 (25.8%)		349 (29.9%)	1749 (26.4%)	
2.01-3	1631 (15.3%)	740 (16.3%)	468 (14.4%)		193 (16.5%)	1015 (15.3%)	
3.01-4	1146 (10.8%)	516 (11.4%)	365 (11.2%)		137 (11.7%)	744 (11.2%)	
4.01-5	2599 (24.4%)	1143 (25.2%)	830 (25.5%)		256 (21.9%)	1718 (25.9%)	

[a]The Wilcoxon rank sum test with continuity correction was performed to assess for statistical significance between the age distributions of participants with and without MASLD and, similarly, for participants with and without Hepatic Fibrosis.
[b]The Chi-Square test was used to analyze categorical data, including Sex, Race, and Income to Poverty Ratio to assess for statistical differences between these variables and MASLD or Hepatic Fibrosis status (with vs without steatosis/stiffness).
[c]These N values represent patients who had complete transient liver elastography data (CAP™ score for steatosis; median liver stiffness for stiffness). In total, there were approximately 9700 patients with complete liver scan data.

rest of the biomarkers being insignificant. The results are available in Table 4.

Similarly for steatosis outcome, variable importance analysis was performed with exclusion of variables with the lowest % IncMSE values. Older age increased the risk of having steatosis (OR = 1.024, $P < .001$). Neither sex was at increased risk ($P = .200$) to have steatosis. Non-Hispanic Black patients were less likely (OR = .466) to have steatosis than other races ($P < .001$). Metabolic profile variables were significantly elevated for steatosis except HbA1c, with BMI (OR = 1.170, $P < .001$) and insulin (OR = 1.033, $P < .001$) showing the strongest significance. Lipid profile variables were not significant for steatosis. Among the standard biochemistry profile variables in steatosis, ALT (2.9%, $P < .001$) and UA (10.9%, $P = .016$) were more likely to be elevated and BUN (4.7%, $P < .001$) to be decreased. The results are available in Table 5.

Model Prediction Performance

The steatosis logistic regression model displayed an overall accuracy of 79.59% (95% CI [0.7812, 0.8099]; $P < .001$). The detection rate of true positives was 72.14% while the detection rate of true negatives was 84.65%. The positive predictive value was 76.16% while the negative predictive value was 81.71%. The stiffness logistic regression model displayed an overall accuracy of 86.07% (95% CI [0.8479, 0.8727]; $P < .001$). The detection rate of true positives was 98.01% while the detection rate of true negatives was 20.92%. The positive predictive value was 87.12% while the negative predictive value was 65.79%. The results are available in Tables 6 and 7.

Sensitivity Analysis

Sensitivity analysis using XGBoost and Recursive Feature Elimination for feature selection was performed for the steatosis and stiffness models. XGBoost had a model prediction performance accuracy of 78.48% (95% CI [0.7625, 0.7909]; $P < .001$) with a specificity of 83.89% and sensitivity of 70.54% for the steatosis model. The positive predictive value was 74.85% and the negative predictive value was 80.72%. For the stiffness model, XGBoost had an accuracy of 85.81% (95% CI [0.8453, 0.8702]; $P < .001$) with a specificity of 19.25% and sensitivity of 98.01%. The positive predictive value was 86.88% and the negative predictive was 63.89%.

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Table 3. Comparison of means among potential biomarker variables

Total (pre-PSM) (N = 12 471)			Steatosis (N = 2714)					Stiffness (N = 988)				
			With hepatic steatosis (MASLD)		without hepatic steatosis (normal)		*P* value	With hepatic stiffness (hepatic fibrosis)		Without hepatic stiffness (normal)		*P* value
	Mean CAP™ score (dB/m)	Mean stiffness (kPa)	Mean CAP™ score (dB/m)					Mean stiffness (kPa)				
	257.14	5.81	304.67		197.28		<.001	11.94		4.74		<.001
Demographics												
Sex	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	—	**Male**	**Female**	**Male**	**Female**	—
	48.3%	51.7%	50.1%	49.9%	45.6%	54.4%	—	54.3%	45.7%	54.3%	45.7%	—
Age (years)	38.34		58.78		36.08		<.001	49.42		49.42		1.00
Income to poverty ratio	2.46		2.71		2.58		.043	2.41		2.48		.451
Metabolic profile												
A1c (%)	5.77		6.23		5.45		<.001	6.34		5.80		<.001
PFG (mg/dL)	111.24		124.09		100.26		<.001	128.12		111.15		<.001
BMI (kg/m²)	27.60		31.98		24.81		<.001	35.68		28.60		<.001
Insulin (μU/mL)	15.05		17.74		9.54		<.001	24.87		12.49		<.001
Lipid Profile												
Total cholesterol (mg/dL)	176.72		188.10		172.10		<.001	171.76		182.35		<.001
HDL-cholesterol (mmol/L)	53.11		50.99		57.25		<.001	48.30		54.18		<.001
LDL-cholesterol (mg/dL)	105.11		112.31		99.98		<.001	100.35		109.08		<.001
Triglycerides (mg/dL)	103.92		123.92		74.37		<.001	115.53		95.43		<.001
Standard biochemistry profile												
ALT (U/L)	20.77		23.23		16.78		<.001	27.11		20.48		<.001
AST (U/L)	21.04		21.38		19.99		.002	24.45		20.54		<.001
ALP (IU/L)	90.25		81.47		93.54		<.001	91.20		83.97		.007
Albumin (g/dL)	4.09		4.00		4.13		<.001	3.97		4.06		<.001
BUN (mg/dL)	14.44		15.82		13.58		<.001	15.60		14.77		.051
GGT (IU/L)	27.93		33.48		21.98		<.001	44.76		27.15		.001
Iron (μg/dL)	84.90		89.22		90.01		.570	84.89		90.75		.009
LDH (IU/L)	159.17		160.46		154.74		<.001	165.52		156.83		.001
Total bilirubin (mg/dL)	0.45		0.50		0.51		.417	0.49		0.49		.940
Total protein (g/dL)	7.17		7.11		7.17		<.001	7.14		7.12		.411
Uric acid (mg/dL)	5.31		5.70		4.98		<.001	5.91		5.42		<.001

Abbreviations: A1c, glycated hemoglobin; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; BUN, blood urea nitrogen; CAP, controlled attenuation parameter; GGT, gamma-glutamyl transferase; HDL, high-density lipoprotein; LDL, low-density lipoprotein; LDH, lactate dehydrogenase; MASLD, metabolic dysfunction–associated steatotic liver disease; PFG, plasma fasting glucose.

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Table 4. Summary of stiffness logistic regression model

Characteristic	Category	Adjusted odds ratio	95% CI	*P* value
Demographics	Age (years)	1.008	[1.001, 1.015]	.023
Sex	Male	Reference		
	Female	0.651	[0.511, 0.828]	.001
Metabolic profile	BMI (kg/m^2)	1.106	[1.089, 1.123]	<.001
	A1c (%)	1.172	[0.988, 1.388]	.068
	PFG (mg/dL)	1.002	[0.997, 1.007]	.400
	Insulin (μU/mL)	1.010	[1.004, 1.017]	.002
Lipid profile	Total cholesterol (mg/dL)	0.993	[0.985, 1.001]	.115
	LDL-cholesterol (mg/dL)	1.000	[0.991, 1.010]	.925
Standard biochemistry profile	ALT (U/L)	1.002	[0.994, 1.011]	.618
	AST (U/L)	1.017	[1.004, 1.030]	.007
	GGT (IU/L)	1.006	[1.002, 1.009]	<.001
	Blood urea nitrogen (mg/dL)	1.024	[1.004, 1.044]	.017
Inflammatory marker	hs-CRP (mg/L)	1.006	[0.983, 1.018]	.393

Abbreviations: A1c, glycated hemoglobin; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; GGT, gamma-glutamyl transferase; hs-CRP, high-sensitivity C-reactive protein; LDL, low-density lipoprotein; PFG, plasma fasting glucose.

Using Recursive Feature Elimination, the stiffness model showed a prediction performance accuracy of 85.90% (95% CI [0.8463, 0.8711]; $P < .001$) with a sensitivity of 97.93%, specificity of 20.29%, positive predictive value of 87.02%, and negative predictive value of 64.24%. The steatosis model had a model prediction performance similar to Random Forest. The model prediction performances comparison among the 3 methods can be found in Table 6 for the stiffness model and Table 7 for the steatosis model.

Assessing for Model Multicollinearity

Multicollinearity was addressed through post hoc variance inflation factor (VIF) calculations. In the steatosis models, the triglycerides, HDL-cholesterol, LDL-cholesterol, and total cholesterol variables produced VIF values greater than 10. The VIF values of these variables for the steatosis logistic regression model are 34.761 (triglycerides), 52.570 (HDL-cholesterol), 120.315 (LDL-cholesterol), and 138.545 (total cholesterol). In the stiffness model none of the variables produced VIF values greater than 10. Tables 8 and 9 display these values in detail.

Discussion

Our paper is the first to use patient survey data to link numerous biomarkers to MASLD and fibrotic liver disease states identified through liver transient elastography measurements (FibroScan). Although several of these biomarkers have been studied individually, we identified biomarkers thought to be linked to MASLD and hepatic fibrosis and created logistic regression models with hepatic steatosis and stiffness data. The combination of biomarkers for MASLD and hepatic fibrosis diagnosis has been previously analyzed. In 2001, Dixon et al determined that an elevated index of insulin resistance, systemic hypertension, and raised ALT levels were each risk factors for MASH when any 2 or all 3 variables were combined (23, 35). Furthermore, Palekar et al differentiated simple steatosis from MASH with a combination of risk factors, including age > 50 years, female gender, AST 45 IU/L, BMI 30 kg/m^2, AST/ALT ratio ≥ 0.80, and hyaluronic acid ≥ 55 mcg/L. when 3 or more risk factors were combined (23, 36). Although results were promising, both studies investigated a limited number of biomarkers and relied on liver biopsy results. Our study, however, provides insight into the associations of a diverse array of biomarkers with hepatic steatosis and stiffness levels assessed by FibroScan, which may be replicated in clinical practice to enhance patient outcomes and identify those at risk for MASLD and hepatic fibrosis at an earlier timeframe.

In 1:1 case to control age- and sex-matched data, we found a strong correlation between both increasing liver steatosis and stiffness and known CVD and MetS risk factors, including lipid profile components, BMI, HbA1c, and PFG (37). Our 2 logistic regression models, each using potential biomarkers for MASLD and hepatic fibrosis, showed moderate accuracy in predicting hepatic steatosis and stiffness, respectively. The MASLD model displayed high specificity (84.65%), indicating it would better serve in prediction of true negative cases of MASLD. The performance of our model is comparable to other well-known scores to calculate people at risk of developing MASLD like the Hepatic Steatosis Index (specificity 92.4%) (38) and Dallas Steatosis Index (DSI) (specificity 90%) (39). The DSI has been more validated, but it only exhibits a high specificity (>90%) when risk threshold is >50%. At lower risk thresholds (40% or lower), the DSI actually exhibits worse specificity (83% vs 85%) than our model, with specificity decreasing as probability decreases (39). The hepatic stiffness model exhibited high sensitivity (98.01%), which indicates its ability to predict true positive cases of hepatic fibrosis more accurately. The performance of hepatic stiffness model was compared to Fibrosis-4 index (FIB-4), as a study showed that it was superior to other known indices for predicting hepatic fibrosis (40). The accuracy of our model (86.07%) was comparable to that of FIB-4 (accuracy 89%); however, it demonstrated much higher sensitivity, suggesting that it would be a more effective screening tool (40): a negative score from our model would help patients avoid further testing. Furthermore, our model has a positive predictive value of 87% compared to 75% seen in FIB-4. This means that our model can aid in avoiding an invasive liver biopsy in positive cases as well.

Given the high specificity of our model for predicting MASLD in low-risk patients, our model could be used to confirm MASLD, preventing the need for liver biopsy. If clinical

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Table 5. Summary of steatosis logistic regression model

Characteristic	Category	Adjusted odds ratio	95% CI	*P* value
Demographics	Age (years)	1.024	[1.018, 1.030]	<.001
Sex	Male	Reference		
	Female	0.857	[0.677, 1.085]	.200
Race	Mexican American	Reference		
	Other Hispanic	0.811	[0.557, 1.180]	.273
	Non-Hispanic White	0.996	[0.734, 1.349]	.979
	Non-Hispanic Black	0.466	[0.334, 0.649]	<.001
	Non-Hispanic Asian	1.197	[0.832, 1.721]	.332
	Other Race/Multi-Racial	0.956	[0.603, 1.518]	.849
	Other Hispanic	0.811	[0.557, 1.180]	.273
Metabolic profile	BMI (kg/m^2)	1.170	[1.146, 1.195]	<.001
	A1c (%)	1.027	[0.843, 1.256]	.796
	PFG (mg/dL)	1.007	[1.000, 1.013]	.039
	Insulin (μU/mL)	1.033	[1.020, 1.047]	<.001
Lipid profile	Total cholesterol (mg/dL)	1.141	[0.828, 1.573]	.420
	LDL-cholesterol (mg/dL)	0.878	[0.637, 1.210]	.427
	HDL-cholesterol (mmol/L)	0.875	[0.634, 1.206]	.414
	Triglycerides (mg/dL)	0.980	[0.919, 1.045]	.534
Standard biochemistry profile	ALT (U/L)	1.029	[1.016, 1.043]	<.001
	AST (U/L)	0.989	[0.972, 1.005]	.170
	ALP (IU/L)	1.001	[0.999, 1.003]	.299
	Albumin (g/dL)	1.246	[0.885, 1.752]	.207
	GGT (IU/L)	0.999	[0.996, 1.001]	.166
	LDH (IU/L)	1.000	[0.997, 1.003]	.942
	Iron (μg/dL)	0.998	[0.995, 1.001]	.166
	Total bilirubin (mg/dL)	0.845	[0.582, 1.221]	.375
	Blood urea nitrogen (mg/dL)	0.953	[0.934, 0.972]	<.001
	Uric acid (mg/dL)	1.109	[1.020, 1.206]	.016
Inflammatory marker	hs-CRP (mg/L)	0.999	[0.985, 1.013]	.839

Abbreviations: A1c, glycated hemoglobin; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; GGT, gamma-glutamyl transferase; HDL, high-density lipoprotein; hs-CRP, high-sensitivity C-reactive protein; LDH, lactate dehydrogenase; LDL, low-density lipoprotein; PFG, plasma fasting glucose.

Table 6. Model prediction performance of stiffness logistic regression model

Statistical variable	Random forest	XGBoost	Recursive feature elimination
Accuracy (95% CI)	86.07% (84.79-87.27%)	85.81% (84.53-87.02%)	85.90% (84.63-87.11%)
***P* value**	<0.001	<0.001	<0.001
Sensitivity	98.01%	98.01%	97.93%
Specificity	20.92%	19.25%	20.29%
Positive predictive value	87.12%	86.88%	87.02%
Negative predictive value	65.79%	63.89%	64.24%
Prevalence	84.51%	84.51%	84.51%
Detection rate	82.83%	82.83%	82.76%
Detection prevalence	95.07%	95.33%	95.11%
Balanced accuracy	59.46%	58.63%	59.11%

suspicion is not high enough and biopsy is still required, our model has a high positive predictive value, and, therefore, still limits unnecessary biopsies. Our model has incredibly increased sensitivity for hepatic fibrosis, indicating its potential use as a screening tool, although patients would still require further confirmation of fibrosis. However, given its sensitivity of 98%, it can help rule out hepatic fibrosis in patients for which there is low clinical suspicion (again avoiding unwarranted biopsies).

Also, to further validate our stiffness and steatosis models for feature selection obtained using Random Forest method, we performed a sensitivity analysis using different methods for feature selection including XGBoost and Recursive Feature Elimination. It showed that the Random Forest method of feature selection had the best model prediction performance for the stiffness model with an accuracy of 86.07% compared to XGBoost (85.81%) and Recursive Feature Elimination (85.90%). All the methods had a similar sensitivity of 98%. At the same time, the model prediction for the steatosis models was better for Random Forest compared to XGBoost with higher accuracy (79.59% vs 78.48%) and specificity (84.65% vs 83.89%). Although, Recursive Feature Elimination had a similar performance to Random Forest. These results confirm the validity and robustness of our proposed models and that the best method of feature selection was chosen for the study. This increases confidence in

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Table 7. Model prediction performance of steatosis logistic regression model

Statistical variable	Random forest	XGBoost	Recursive feature elimination
Accuracy (95% CI)	79.59% (78.12-80.99%)	78.48% (76.25-79.09%)	79.59% (78.12-80.99%)
***P* value**	<0.001	<0.001	<0.001
Sensitivity	72.14%	70.54%	72.14%
Specificity	84.65%	83.89%	84.65%
Positive predictive value	76.16%	74.85%	76.16%
Negative predictive value	81.71%	80.72%	81.71%
Prevalence	40.47%	40.47%	40.47%
Detection rate	29.20%	28.55%	29.20%
Detection prevalence	38.33%	38.14%	38.33%
Balanced accuracy	78.39%	77.21%	78.39%

Table 8. Variance inflation factor of stiffness model

Characteristic	Category	Variance inflation factor (VIF)
Demographics	Age (years)	1.447
	Sex	1.216
Metabolic profile	BMI (kg/m^2)	1.296
	A1c (%)	4.263
	PFG (mg/dL)	4.130
	Insulin (μU/mL)	1.157
Lipid profile	Total cholesterol (mg/dL)	7.925
	LDL-cholesterol (mg/dL)	7.829
Standard biochemistry profile	ALT (U/L)	2.729
	AST (U/L)	2.615
	GGT (IU/L)	1.237
	Blood urea nitrogen (mg/dL)	1.283
Inflammatory marker	hs-CRP (mg/L)	1.117

Abbreviations: A1c, glycated hemoglobin; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; GGT, gamma-glutamyl transferase; hs-CRP, high-sensitivity C-reactive protein; LDL, low-density lipoprotein; PFG, plasma fasting glucose.

Table 9. Variance inflation factor of steatosis model

Characteristic	Category	Degree of freedom (Df)	Variance inflation factor (VIF)
Demographics	Age (years)	1	1.377
	Sex	1	1.285
	Race/Ethnicity	5	1.039
Metabolic profile	BMI (kg/m^2)	1	1.236
	A1c (%)	1	1.781
	PFG (mg/dL)	1	1.706
	Insulin (μU/mL)	1	1.145
Lipid profile	Total cholesterol (mg/dL)	1	138.545
	LDL-cholesterol (mg/dL)	1	120.315
	HDL-cholesterol (mmol/L)	1	52.570
	Triglycerides (mg/dL)	1	34.761
Standard biochemistry profile	ALT (U/L)	1	1.587
	AST (U/L)	1	1.796
	ALP (IU/L)	1	1.124
	Albumin (g/dL)	1	1.226
	GGT (IU/L)	1	1.228
	LDH (IU/L)	1	1.206
	Iron (μg/dL)	1	1.206
	Total Bilirubin (mg/dL)	1	1.175
	Blood urea nitrogen (mg/dL)	1	1.180
	Uric acid (mg/dL)	1	1.228
Inflammatory marker	hs-CRP (mg/L)	1	1.060

Abbreviations: A1c, glycated hemoglobin; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; GGT, gamma-glutamyl transferase; HDL, high-density lipoprotein; hs-CRP, high-sensitivity C-reactive protein; LDH, lactate dehydrogenase; LDL, low-density lipoprotein; PFG, plasma fasting glucose.

selecting the biomarkers predicted by Random Forest for the steatosis and stiffness models in detection of early cases of MASLD and hepatic fibrosis using these biomarkers.

Limitations

Some of the limitations in this study relate to the use of FibroScan measurements to define steatosis and stiffness as outcomes. It is also important to note that, in this study, we have not confirmed histologic evidence of MASLD or hepatic fibrosis with liver biopsy. Although the advantages of FibroScan include its safety and noninvasive nature, there are discrepancies between its measurements and liver biopsy, the current gold standard. Substantial evidence has validated transient elastography for the evaluation of fibrosis in conditions like viral hepatitis and liver cirrhosis. Meta-analyses have found moderate sensitivity and specificity for FibroScan's detection of fibrosis. In Geng et al's 2016 meta-analysis of 57 articles with over 10 000 patients, transient elastography detected fibrosis with an overall sensitivity of 81%. Xiao et al's 2017 meta-analysis of 15 FibroScan studies with 3641 patients investigated a narrower focus of fibrotic disease severity. For the diagnosis of significant fibrosis, advanced fibrosis, and cirrhosis, they found pooled sensitivities of 71.6%, 79.0%, and 80.0%, respectively.

Longitudinal data are still necessary for establishing the relationship between CAP and clinical outcomes. Recent findings indicate moderate specificities and sensitivities of CAP in steatosis assessment. In 2020, Zhang et al reported moderate to high sensitivity and specificity ranges in their review of CAP scores in studies that used hepatic histology as reference (sensitivities: ≥S1, ≥S2, and S3 are 60%-75%, 69%-84% and 77%-96%, respectively; specificities: 76%-90%, 75%-88% and 72%-82%, respectively). Similarly, a 2017 meta-analysis of CAP by Karlas et al of 2735 patients involving 19 studies found pooled sensitivities and specificities of 69% and 82% for ≥ S1, 77% and 81% for ≥ S2, and 88% and 78% for S3, respectively. However, of the patients studied with liver disease, only about 20% had MASLD/MASH. Although it has

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been determined that CAP is a more objective tool than standard abdominal ultrasonography in steatosis evaluation, its accuracy must be further studied (9, 41-48).

Related to our logistic regression models, a drawback is the number of variables, which may lead to model misinterpretation. Although our use of a multivariate analysis does mimic the clinical setting where several biomarkers often occur in one patient, this introduces multicollinearity. Multicollinearity occurs when variables used in the regression model are correlated, leading to potential bias when interpreting regression coefficients. To address this, we calculated a post hoc variance inflation factor (VIF) for each of our predictor variables. VIF values greater than 10 are generally considered unfavorable, indicating possible inflation of regression coefficients due to multicollinearity. High VIF values were notably observed for triglycerides, HDL-cholesterol, LDL-cholesterol, and total cholesterol in the steatosis model, which are all inherently related components of the lipid profile. Including individual lipid variables is crucial to elucidating the relationship of each biomarker with hepatic steatosis and stiffness, which may be obscured by aggregating these lipid variables. Additionally, because stronger predictor variables greatly impact the model, variables that are less contributory to the model's accuracy should be interpreted with caution as stronger predictor variables may overshadow their significance. Tables 4 and 5 explicitly indicate which variables were statistically significant in the models created in this study. However, considering methods that reduce the number of variables and eliminate multicollinearity may provide more interpretable and reliable results (49).

Strengths of the Study

The strengths of our study include analyzing data using propensity score matching with a 1:1 case to control for age and sex as well as machine learning tools like Random Forest. Propensity score matching allows for the balancing of covariates without randomization and increases intergroup comparability. Randomized controlled trials produce the highest form of evidence in medical research, however, these studies are not always attainable due to ethical, practicality, safety, and cost issues. In observational studies, when randomization is not possible, propensity score matching can be used to balance covariates between groups to ensure more accurate comparability (50, 51). In our study's initial assessment of potential biomarkers, each patient without steatosis (control) is matched by age and sex to a patient with steatosis (MASLD; CAP $\geq$ 238 dB/m; case). This was additionally performed based upon median stiffness values where case was defined as $\geq$7 kPa (hepatic fibrosis).

The Random Forest algorithm can estimate variable importance by assessing the increase in prediction error when data for a variable is altered while others remain unaltered. This was especially useful for model reduction in our study as a multitude of variables were assessed. Therefore, although our logistic regression model contains a large number of variables, included variables were determined as having the highest variable importance using Random Forest (52).

Lastly, we investigated a variety of biomarkers, all of which are typically clinically accessible. Our models incorporate these biomarkers and could be applied to clinical practice during patient assessment. Patients with concerning steatosis and stiffness results may then undergo invasive liver biopsy, where pathological evaluation can confirm liver disease. By providing the tools for noninvasive and accessible early detection of abnormal liver changes, we aim to reduce the disease burden of MASLD and hepatic fibrosis.

Future Directions

Testing our model with biopsy-confirmed cases of MASLD and hepatic fibrosis would provide further validation of its effectiveness. Future studies should further investigate the relationship between CVD and MetS risk factors and MASLD as there are strong associations between these disease processes. Similarly, a future study could evaluate the effectiveness of a weight loss intervention by including it as a covariate. Additional evidence related to the use of FibroScan in clinical practice is also warranted, especially if future studies utilize these same measurements to define MASLD and hepatic fibrosis. Our model can also be used in a future longitudinal study to assess the change in biomarkers from MASLD to hepatic fibrosis. Lastly, investigation into the exact pathogenic mechanisms underlying MASLD and its association with MetS may also provide guidance toward other biomarkers or avenues for early detection.

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Disclosures

All the authors involved in the study do not have any conflicts of interest to disclose.

Data Availability

The data utilized for the study is an open source publicly available National Health and Nutrition Examination Survey 2017-2020 database available at https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle = 2017-2020 under the section data, documentation, and codebooks. The data was analyzed using the open-source statistical software R.

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